Quiz — 2.2.2 Computational Methods

Name: ____ Date: _ ***Mark:*** ___ / 25

OCR H446 · Component 02 · 2.2.2

Section A — Multiple choice (8 marks)

A1. Which best describes a problem that is **solvable computationally**? A. Any problem a human can describe in words B. A problem with a clear, finite sequence of rule-based steps that can be decomposed C. Only problems that involve numbers D. A problem that always has exactly one solution Your answer: ☐

A2. Decomposition means: A. Hiding unnecessary detail B. Breaking a complex problem into smaller, manageable sub-problems C. Trying options and backing up at dead ends D. Predicting performance with a model Your answer: ☐

A3. Divide and conquer is most directly associated with which technique? A. Linear search B. Recursion (splitting into smaller instances of the same problem) C. Pipelining D. Data mining Your answer: ☐

A4. A satnav giving a fast, **good-enough but not guaranteed optimal** route is using a: A. Heuristic B. Brute-force search C. Backtracking algorithm D. Visualisation Your answer: ☐

A5. Solving a maze by going back to the last junction after hitting a dead end is: A. Abstraction B. Pipelining C. Backtracking D. Performance modelling Your answer: ☐

A6. Searching very large data sets for hidden patterns and correlations is: A. Visualisation B. Data mining C. Decomposition D. Heuristics Your answer: ☐

A7. Splitting a process into stages so different stages work on **different items at the same time** is: A. Backtracking B. Pipelining C. Abstraction D. Divide and conquer Your answer: ☐

A8. The Tube map removing exact distances and showing only line connections is an example of: A. Performance modelling B. Representational abstraction C. Data mining D. Backtracking Your answer: ☐

Section B — Short answer (12 marks)

B1. State **two** characteristics that make a problem solvable computationally. **[2]**

B2. Match each computational method (1–5) to the most appropriate use (A–E). Write the letter next to each number. **[3]**

#	Method		Use
1	Backtracking	A	Predicting how a server copes with load before it is built
2	Data mining	B	Solving Sudoku by undoing a guess at a dead end
3	Heuristics	C	Loyalty-card basket analysis to find buying patterns
4	Performance modelling	D	Representing infection rates as a colour-coded heat map
5	Visualisation	E	A* route-finding giving a quick near-optimal answer

1 → _ ***2*** → *3* → _ ***4*** → 5 → ____

B3. Explain the difference between **abstraction by representation** and **abstraction by generalisation**, giving one example of each. **[2]**

B4. A company wants to model the spread of a delivery route over a road network. Explain how **divide and conquer** could be applied, and state why it pairs naturally with **recursion**. **[2]**

B5. A heuristic is used to solve the Travelling Salesman Problem. Explain **why** a heuristic is used here rather than an exact method. **[2]**

B6. State **one** benefit of **visualisation** when analysing large data sets. **[1]**

Section C — Exam-style question (5 marks)

A games company is writing a program to place 8 queens on a chessboard so that no two attack each other (the N-Queens problem).

(a) Name the computational method best suited to this problem. **[1]**

(b) Describe **how** the chosen method would attempt to find a valid arrangement, including what happens at a **dead end**. **[3]**

(c) State **one** reason this method may still be slow for large boards. **[1]**